Soham Chainani
D20A 49

EXPERIMENT 10

AIM:

Implementation of Hyperledger Fabric Network and Asset Management using Chaincode

THEORY:

Hyperledger Fabric is an open-source, permissioned blockchain platform designed for enterprise
applications. Unlike public blockchains, it allows only authorized participants to join the network. It follows a modular architecture consisting of organizations, peers, an ordering service,
a Membership Service Provider (MSP), and smart contracts known as chaincode.

A Fabric network consists of multiple organizations, each having peers that maintain a copy of the ledger. The ledger contains the blockchain, which stores transaction history, and the world state, which maintains the latest values of assets. The ordering service collects endorsed transactions, orders them into blocks, and distributes the blocks to peers. Peers validate transactions according to endorsement policies before updating the ledger.

Chaincode implements the business logic of a blockchain application. In this experiment, the Hyperledger Fabric test network was configured with two organizations, Org1 and Org2, and a channel named mychannel. The Go-based basic chaincode was deployed to the channel to perform asset-management operations.

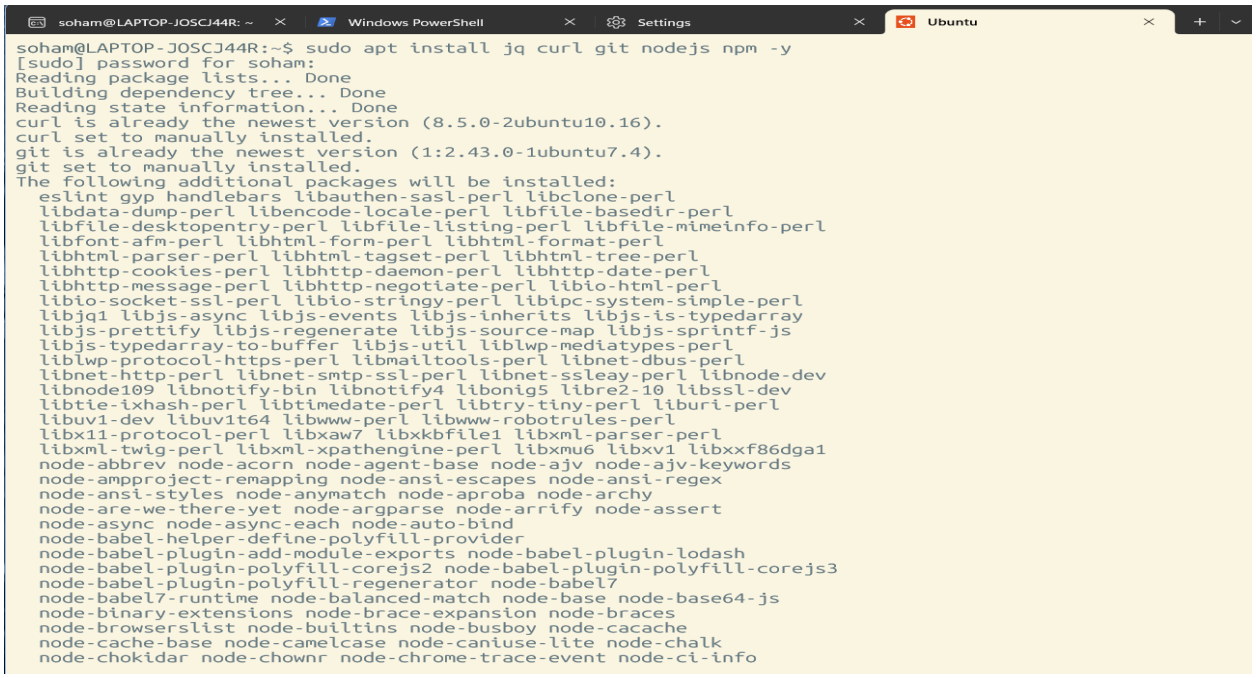
The main asset-management operations are:
1. Create Asset: Adds a new asset with attributes such as asset ID, color, size, owner, and value.
2. Read Asset: Retrieves the details of an existing asset from the ledger.
3. Update Asset: Modifies the attributes of an existing asset.
4. Get All Assets: Retrieves the list of assets stored in the ledger.
5. Transfer Asset: Changes the ownership of an existing asset.

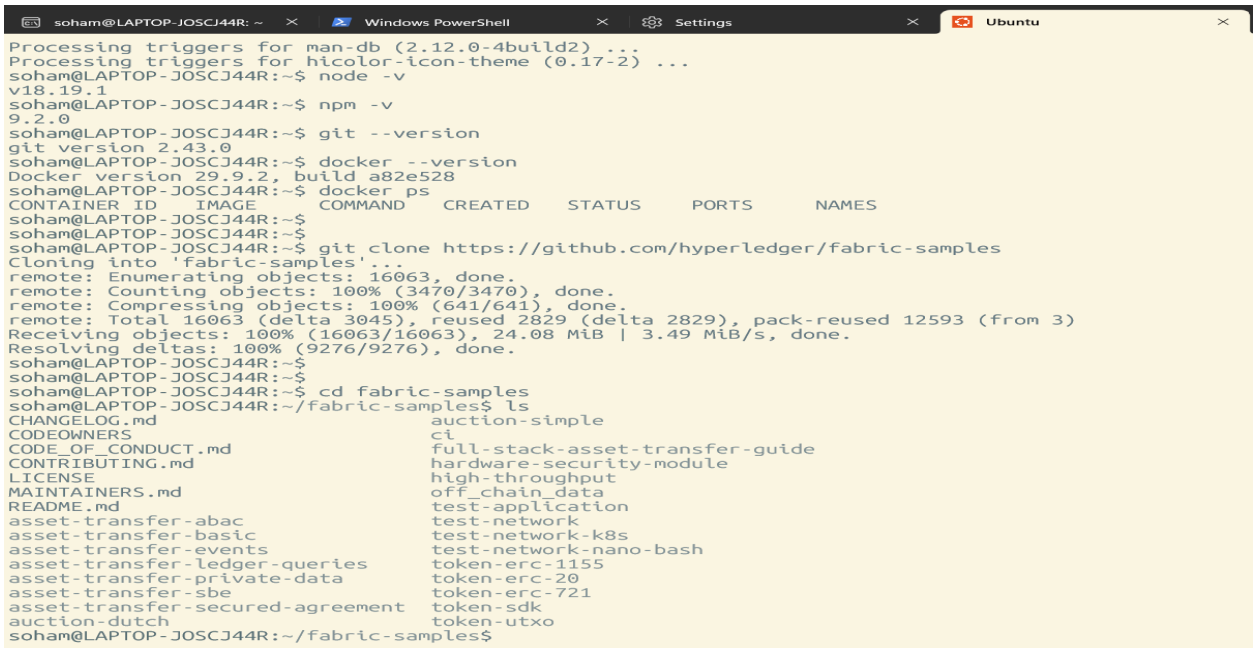
when a client subinits a transaction, the required peers execute the chaincode and provide endorsements. The transaction is then submitted to the ordering service, which orders transactions into blocks. Peers validate the endorsements and transaction status before committing valid transactions to the ledger. In this experiment, the default endorsement policy required endorsements from both organizations, demonstrating how Fabric enforces transaction authorization and validation.

Hyperledger Fabric provides security, transparency, traceability, data consistency, and controlled access. Since transactions are recorded on the blockchain, asset changes can be tracked, making

the platform suitable for supply-chain management, inventory tracking, and enterprise asset management.

CODE AND OUTPUT:



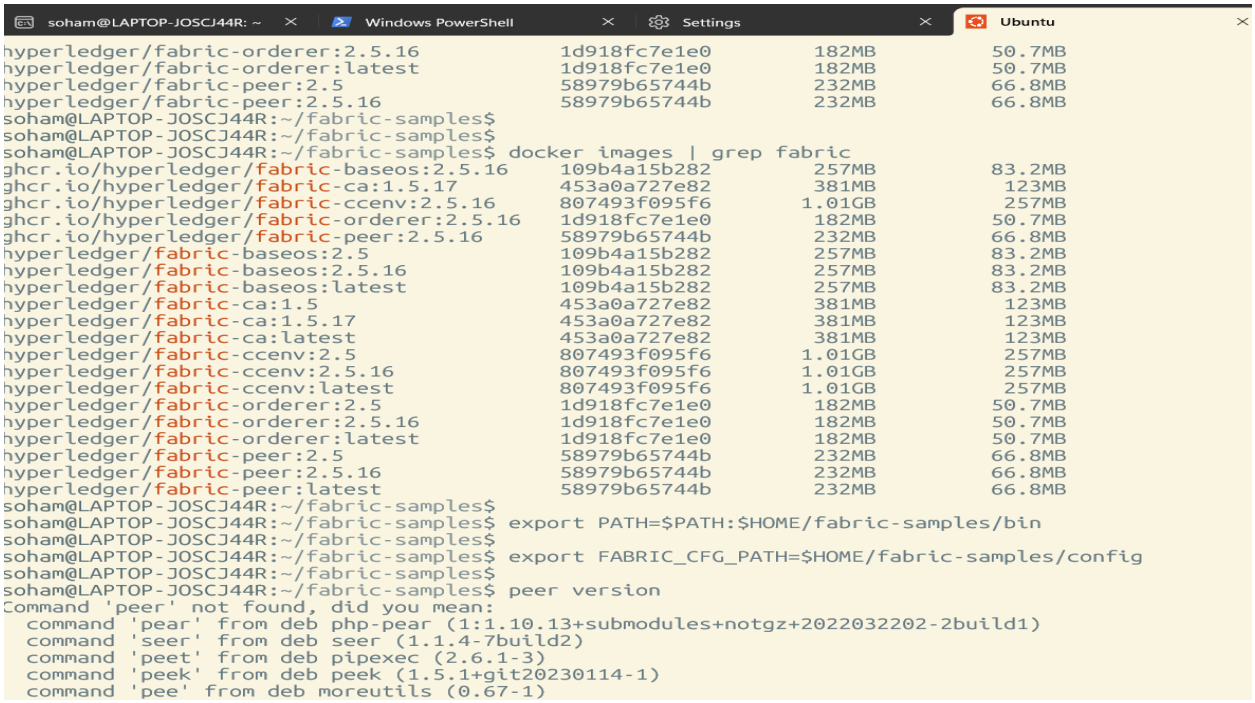


```
soham@LAPTOP-JOSCJ44R:~/fabric-samples$ ~/fabric-samples
-bash: /home/soham/fabric-samples: Is a directory
soham@LAPTOP-JOSCJ44R:~/fabric-samples$ pwd
/home/soham/fabric-samples
soham@LAPTOP-JOSCJ44R:~/fabric-samples$
soham@LAPTOP-JOSCJ44R:~/fabric-samples$ curl -sSLO https://raw.githubuser
content.com/hyperledger/fabric/main/scripts/install-fabric.sh
soham@LAPTOP-JOSCJ44R:~/fabric-samples$
soham@LAPTOP-JOSCJ44R:~/fabric-samples$ chmod +x install-fabric.sh
soham@LAPTOP-JOSCJ44R:~/fabric-samples$
soham@LAPTOP-JOSCJ44R:~/fabric-samples$ ./install-fabric.sh docker

Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccenv baseos
===> Pulling fabric Images
====>  ghcr.io/hyperledger/fabric-peer:2.5.16
2.5.16: Pulling from hyperledger/fabric-peer
6b3c778533c0: Pull complete
f2df6aa6a776: Pull complete
b7fdc54bd9fd: Pull complete
40d16f30db40: Pull complete
cefbeedeeae1: Pull complete
472c4fdce475: Pull complete
092c57c5022d: Download complete
Digest: sha256:58979b65744b5e7f775d5c65af3831d9beb344d8294c6aa82cbae59e7f
7760d6
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-peer:2.5.16
ghcr.io/hyperledger/fabric-peer:2.5.16
====>  ghcr.io/hyperledger/fabric-orderer:2.5.16
2.5.16: Pulling from hyperledger/fabric-orderer
f2d6cf9c8d1b: Pull complete
40d16f30db40: Already exists
15fd4713e88f: Pull complete
7eed68e83b36: Pull complete
dcc2ada4d1af: Pull complete
ad07741a7783: Pull complete
7d6e7f31e9cc: Download complete
Digest: sha256:1d918fc7e1e05b28a9a2f91a61f6435d058f156ef97b5005271de66fbf
1370f1
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-orderer:2.5
.16
```

```
Digest: sha256:453a0a727e82c4960b797e379adc231e853ee23ae7526db53afef77b50
74117e
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-ca:1.5.17
ghcr.io/hyperledger/fabric-ca:1.5.17
===> List out hyperledger images
ghcr.io/hyperledger/fabric-baseos:2.5.16     109b4a15b282         257MB
      83.2MB
ghcr.io/hyperledger/fabric-ca:1.5.17         453a0a727e82         381MB
       123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.16      807493f095f6        1.01GB
       257MB
ghcr.io/hyperledger/fabric-orderer:2.5.16    1d918fc7e1e0         182MB
      50.7MB
ghcr.io/hyperledger/fabric-peer:2.5.16       58979b65744b         232MB
      66.8MB
hyperledger/fabric-baseos:2.5                109b4a15b282         257MB
      83.2MB
hyperledger/fabric-baseos:2.5.16             109b4a15b282         257MB
      83.2MB
hyperledger/fabric-baseos:latest             109b4a15b282         257MB
      83.2MB
hyperledger/fabric-ca:1.5                    453a0a727e82         381MB
       123MB
hyperledger/fabric-ca:1.5.17                 453a0a727e82         381MB
       123MB
hyperledger/fabric-ca:latest                 453a0a727e82         381MB
       123MB
hyperledger/fabric-ccenv:2.5                 807493f095f6        1.01GB
       257MB
hyperledger/fabric-ccenv:2.5.16              807493f095f6        1.01GB
       257MB
hyperledger/fabric-ccenv:latest              807493f095f6        1.01GB
       257MB
hyperledger/fabric-orderer:2.5               1d918fc7e1e0         182MB
      50.7MB
hyperledger/fabric-orderer:2.5.16            1d918fc7e1e0         182MB
      50.7MB
hyperledger/fabric-orderer:latest            1d918fc7e1e0         182MB
      50.7MB
hyperledger/fabric-peer:2.5                  58979b65744b         232MB
      66.8MB
hyperledger/fabric-peer:2.5.16               58979b65744b         232MB
```




```
soham@LAPTOP-JOSCJ44R:~/fabric-samples$ export PATH=$PATH:$HOME/fabric-sa
mples/bin
peer version

peer:
 Version: v2.5.16
 Commit SHA: f871cf9
 Go version: go1.26.4
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger

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```

```
soham@LAPTOP-JOSCJ44R:~/fabric-samples$ cd ~/fabric-samples/test-network
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ ./network.sh up createChannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb with crypto from 'cryptogen'
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
/home/soham/fabric-samples/test-network/../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
[+] up 7/7
 ✔ Network fabric_test                      Created                    0.1s
 ✔ Volume compose_peer0.org2.example.com    Created                    0.0s
 ✔ Volume compose_orderer.example.com       Created                    0.0s
 ✔ Volume compose_peer0.org1.example.com    Created                    0.0s
 ✔ Container peer0.org2.example.com         Started                    1.4s
 ✔ Container peer0.org1.example.com         Started                    1.9s
 ✔ Container orderer.example.com            Started                    2.5s
CONTAINER ID   IMAGE                                COMMAND                  CREATED         STATUS                       PORTS                                                
```

```
Chaincode initialization is not required
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ export PATH=$PATH:$HOME/fabric-samples/bin
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ export CORE_PEER_TLS_ENABLED=true
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ export CORE_PEER_LOCALMSPID="Org1MSP"
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganization
s/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/or
g1.example.com/users/Admin@org1.example.com/msp
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ export CORE_PEER_ADDRESS=localhost:7051
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
```

```
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com
-cert.pem" \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
-c '{"function":"InitLedger","Args":[]}'
2026-11-05 19:33:00.746 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"},{"AppraisedValue":400,"Color":"red","ID":"asset2","Owne
r":"Brad","Size":5,"docType":"asset"},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"},{"AppraisedValu
e":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"},{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","
Size":15,"docType":"asset"},{"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset5"]}'
{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ CreateAsset(ID, Color, Size, Owner, AppraisedValue)
-bash: syntax error near unexpected token `ID,'
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com
-cert.pem" \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
--peerAddresses localhost:9051 \
```

```
2026-11-05 19:33:45.898 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"{\"ID\":\"asset7
\",\"Color\":\"blue\",\"Size\":20,\"Owner\":\"Manav\",\"AppraisedValue\":500}"
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com
-cert.pem" \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles "$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt" \
-c '{"function":"TransferAsset","Args":["asset7","Ravi"]}'
2026-11-05 19:33:58.794 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"Manav"
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset7"]}'
{"AppraisedValue":500,"Color":"blue","ID":"asset7","Owner":"Ravi","Size":20}
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$ ./network.sh down
Using docker and docker-compose
Stopping network
[+] down 7/7
 ✔Container orderer.example.com         Removed                    1.0s
 ✔Container peer0.org1.example.com      Removed                    2.1s
 ✔Container peer0.org2.example.com      Removed                    1.7s
 ✔Volume compose_peer0.org1.example.com Removed                    0.1s
 ✔Volume compose_orderer.example.com    Removed                    0.1s
 ✔Volume compose_peer0.org2.example.com Removed                    0.1s
 ✔Network fabric_test                   Removed                    0.4s
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc1497615f5e33a8bfa908e0fd
3555f864cfa1398b52d341cb:latest
Deleted: sha256:b8e4bc542bd8895275b2bf6862f504feac4cd70aabe66d31ef5c14dc6bc11ef5
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629-d3cf9b1f4b9747cc8433ac0e74bfaeade1cf0b97
f5c1163835f604aa885d60dc:latest
Deleted: sha256:e6d91b09b92bba92e156ae5d8d39ab7e62fdcb28b89b73b93fc9520cb006654b
```

```
Removing remaining containers
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc1497615f5e33a8bfa908e0fd
3555f864cfa1398b52d341cb:latest
Deleted: sha256:b8e4bc542bd8895275b2bf6862f504feac4cd70aabe66d31ef5c14dc6bc11ef5
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629-d3cf9b1f4b9747cc8433ac0e74bfaeade1cf0b97
f5c1163835f604aa885d60dc:latest
Deleted: sha256:e6d91b09b92bba92e156ae5d8d39ab7e62fdcb28b89b73b93fc9520cb006654b
Unable to find image 'busybox:latest' locally
latest: Pulling from library/busybox
37bb94b0940b: Pull complete
46500c34ad0c: Download complete
Digest: sha256:a46257e06c82ad85242d0d38077d0a52d9896389ca60a1a59d3463db586e5fa4
Status: Downloaded newer image for busybox:latest
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
soham@LAPTOP-JOSCJ44R:~/fabric-samples/test-network$
```

Result

The Hyperledger Fabric network was successfully implemented. JavaScript chaincode was deployed, and asset operations such as creation, querying, and ownership transfer were performed successfully. The asset was initially created with owner name Manav and subsequently transferred to Ravi.

Conclusion

This experiment demonstrated the implementation of a blockchain network using Hyperledger Fabric. It provided practical understanding of chaincode deployment, ledger initialization, asset creation, querying, and ownership transfer. It also demonstrated how blockchain maintains a secure and traceable record of asset transactions.